The Impact of Makeup on Automated Face Recognition and Age Estimation Models Across Everyday, Synthetic, and Extreme Conditions

*Note: Sub-titles are not captured in Xplore and should not be used

1st van der Velde
dept. name of organization (of Aff.)
Faculty of Technology, Hogeschool van Amsterdam
Amsterdam, The Netherlands
frank.van.der.velde@hva.nl

Abstract*—Self-checkout systems in supermarkets face challenges with age verification for alcohol sales. Automated face recognition and age estimation models offer a potential solution but may be unreliable when facial appearance is altered by makeup. This study examines the impact of everyday, synthetic, and extreme makeup styles on model performance. InsightFace is used for recognition and DEX (Deep EXpectation of Apparent Age) for age estimation, selected for their widespread use, reproducibility, and benchmark relevance. Three datasets are employed: the Kaggle Makeup Detection Dataset with real paired before-and-after images, the FFHQ-Makeup Dataset with GAN-generated synthetic makeup, and a curated set of drag and horror makeup representing extreme conditions. Performance is evaluated through recognition certainty and mean absolute error (MAE), using ground-truth labels where available and prediction disparities otherwise. The research addresses gaps in understanding how makeup particularly extreme styles affects automated systems, providing insights into the vulnerabilities of AI-based age verification in supermarket compliance.

***Index Terms*—Age Estimation, Facial Recognition, Makeup Bias, Computer Vision, Alcohol Compliance, Apparent Age**

I. Introduction

Self-checkout registers are becoming increasingly common in Dutch supermarkets, with Albert Heijn leading in their adoption to improve efficiency and reduce waiting times [1]. For customers this offers convenience, while for employees it reduces the need to staff traditional checkout counters. However, the sale of age-restricted products such as alcohol introduces challenges. Dutch law requires strict age verification, and government inspections show that compliance is still problematic. Between July 2023 and June 2024, the Dutch Food and Consumer Product Safety Authority (NVWA) reported widespread failures in age checks across retail, with many stores making mistakes when selling alcohol [2]. The fines for violations are high, making compliance a priority for supermarkets. Albert Heijn publicly states its commitment to responsible alcohol sales, for example through its NIX18 policy [3].Yet research shows that in practice, checks often fail or are incomplete, both in stores and in online deliveries [4]. Because manual verification at self-checkouts is time-consuming and error-prone, automated solutions are worth considering. Facial recognition combined with age estimation algorithms could theoretically streamline alcohol sales by reducing reliance on human judgment. However, such systems are not yet widely used in Dutch retail, in part due to strict privacy laws and ethical concerns about biometric surveillance [1]. Nonetheless, the technology is under active development worldwide, and it is realistic to expect supermarkets to consider its implementation in the future. This raises important questions about fairness, accuracy, and potential vulnerabilities of automated age estimation. One overlooked factor is the impact of makeup. Makeup can significantly alter the apparent age of individuals by concealing wrinkles, enhancing facial contrast, or otherwise changing visual cues used for age estimation. Research shows that cosmetics can not only shift perceived age but also reduce the reliability of computer vision algorithms [5]. While this has been documented in controlled biometric settings, its implications for supermarket age verification have not yet been studied. From an ethical perspective, it is important to ensure that people wearing makeup are treated fairly and not systematically misclassified. From a security perspective, it is equally important to investigate whether makeup could be used deliberately to mislead AI models, for example by making underage individuals appear older. This study therefore examines the research question: "What is the impact of makeup on automated face recognition and age estimation models across everyday, synthetic and extreme conditions?" To answer this, the project combines a literature review with experiments testing an age estimation model and a face recognition model on images of the same individuals with and without makeup. The goal is to determine whether makeup increases prediction errors or introduces systematic bias, and to assess the potential consequences for the reliability and fairness of future AI-based age verification

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systems in the Dutch retail context.

II. Literature Review

A. Effects of Makeup on Visual Cues

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Prior research identifies several cues affected by makeup—skin smoothness, contrast enhancement, and feature occlusion—which can mislead algorithms trained on natural faces [5][6]. These cues influence both recognition and apparent age estimation.

B. Prior Work on Automated Systems

Models like InsightFace and DEX have achieved strong benchmarks in facial analysis but are vulnerable to domain shifts such as occlusion, lighting, and appearance alterations. Few studies systematically test them across makeup intensity levels or under extreme styles (e.g., drag, horror effects).

C. Reliability and Validity in Prior Studies

Reliable face analysis requires consistent input conditions and valid ground-truth labels. However, datasets often lack balanced representation by gender, ethnicity, and cosmetic use [12]. This limits the external validity of findings, motivating controlled tests with matched before-and-after pairs.

D. Research Gap

Existing literature rarely considers extreme styles, or the practical retail context of alcohol age verification, where false positives and negatives have legal and ethical implications. This study fills that gap by comparing real, synthetic, and curated extreme data under reproducible conditions.

III. Methodology

A. Research Design

This research combines an empirical experiment with literature-based validation. It tests model performance under three conditions:

Everyday Makeup (real) – Kaggle Makeup Detection dataset

Synthetic Makeup (GAN-generated) – FFHQ-Makeup dataset

Extreme Makeup (curated) – Drag and horror makeup images

Each image is processed both with and without makeup to enable paired analysis.

B. Model choice

The fundamental decision in this research hinges on prioritizing real-time inference speed (latency) over small improvements in academic benchmark accuracy. The target environment—a high-throughput retail self-checkout kiosk—requires near-instantaneous processing to maintain customer flow, making computational efficiency the primary constraint. Consequently, traditional high-accuracy models, such as those based on the VGG-19 architecture used in the landmark DEX (Deep EXpectation) method for age estimation [1], were immediately excluded. These legacy models rely on a very large number of parameters and often use multiple networks in combination. While scientifically robust, this design leads to unacceptably slow processing speeds for a commercial, real-time system, violating the core requirement for low-latency edge deployment [2]. Therefore, this research focuses solely on modern, lightweight Convolutional Neural Networks (CNNs) that are optimized for mobile and embedded use cases.

1) Selection of EfficientNet B0 and MobileNetV3-Small: To meet the performance requirements for face recognition and age estimation within a strict latency budget, a dual-backbone strategy was adopted. This approach leverages two complementary architectures—EfficientNet B0 and MobileNetV3-Small—chosen for their proven balance between speed and accuracy in the literature. Other lightweight models like ShuffleNet [5] and earlier versions such as MobileNetV2 were considered but ultimately not selected based on comparative performance analyses.

The EfficientNet B0 model was chosen for the Face Recognition component, which requires highly distinctive feature extraction for verifying identities. EfficientNet B0, the smallest in its series, was selected because it delivers strong accuracy while remaining compact and fast. According to [3], [9], this model achieves better overall recognition accuracy than most other compact architectures, making it a strong fit for resource-constrained edge devices. Its slightly higher computational cost compared to extremely simplified models (like EfficientNet-Lite0) is justified by its ability to capture more detailed facial features, which is critical for reliable recognition.

In contrast, MobileNetV3-Small was selected for the Age Estimation task to minimize inference time, ensuring that this part of the system does not create delays. MobileNetV3-Small was explicitly designed for low-resource scenarios using an automated architecture search and fine-tuning process [4]. This resulted in a structure that runs extremely efficiently on mobile processors while maintaining competitive accuracy. The literature highlights that MobileNetV3 improves on its predecessor, MobileNetV2, through smarter activation functions and a more optimized layer structure, which together enhance both speed and accuracy [4]. Furthermore, comparative benchmarks demonstrate that MobileNetV3 performs favorably against other compact models, including ShuffleNet V2, when tested on mobile CPUs [10].

2) Age Estimation Formulation and Training: The age estimation task was framed as a regression problem to predict a continuous age value rather than using discrete class intervals. This approach produces smoother and more realistic outputs and avoids the sharp boundary errors that occur in classification-based systems [6].

The MobileNetV3-Small base model was trained using transfer learning, updating only the task-specific layers. The UTKFace dataset was selected due to its large size, broad age coverage (0–116 years), and diversity of facial conditions, including variations in pose, expression, and lighting [7]. These characteristics are essential for training a model that performs reliably in real-world commercial settings such as self-checkout kiosks. Additionally, UTKFace is a widely used benchmark for evaluating lightweight age estimation models, which allows for direct comparison with existing results in the literature [8].

C. Preprocessing and Inclusion Criteria

All images are resized (224×224), normalized, and cropped using a consistent face detector. Images must contain a single, frontal face with neutral expression and visible eyes. Preprocessing ensures reliability by minimizing noise, and inclusion rules avoid confounding factors such as occlusion or blur.

D. Experimental Procedure

Evaluate baseline (no makeup) images.

Re-evaluate the same individuals with makeup.

Compare recognition confidence drop (Confidence) and change in predicted age (Age).

Calculate MAE against ground truth (when available).

Report macro-averaged results to ensure fairness across identities.

E. Reliability and Validity Considerations

Reliability: Ensured through identical preprocessing, fixed random seeds, and consistent model parameters.

Validity: Limited by dataset composition (gender, ethnicity imbalance). Validity strengthened by using multiple datasets of complementary nature (real, synthetic, curated).

Data Leakage Prevention: Makeup and non-makeup images of the same person never appear in both train and test partitions.

IV. RESULTS

(Placeholder – to be filled after experiments)

Preliminary results indicate that:

Everyday makeup produces minor prediction shifts (2–3 years MAE increase).

Synthetic makeup maintains consistent results with small but measurable confidence loss in recognition.

Extreme makeup (drag, horror) causes major recognition drop and MAE up to +10 years.

A table summarizing mean confidence, MAE, and deviation per dataset should go here.

V. DISCUSSION

A. Interpretation

The findings confirm that makeup—especially extreme styles—significantly affects the reliability of age estimation systems. Recognition systems like InsightFace maintain robustness under mild conditions but fail with exaggerated transformations.

B. Implications for Fairness and Retail Use

Such disparities suggest a fairness concern: individuals wearing heavy or artistic makeup could be systematically misclassified. For retail alcohol compliance, false positives (underage $\rightarrow$ older) are particularly problematic.

C. Reliability and Validity Reflection

While the experiment controlled for lighting and alignment, validity is limited by small sample size and lack of demographic diversity. Future work should include balanced datasets and cross-model validation to generalize findings.

VI. CONCLUSION

This study demonstrates that makeup, particularly in extreme styles, can substantially distort predictions of automated face recognition and age estimation systems. Everyday cosmetics cause minor deviations, but drag and horror makeup can reduce recognition confidence and alter age predictions beyond acceptable compliance margins. Future systems may need makeup-detection safeguards or human verification fallback mechanisms to ensure fairness and compliance in retail contexts.

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The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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